

NICK BUTTS

SENIOR STAFF SOFTWARE
DESIGN ENGINEER

PLOWS TO NEURONS:

THE ROLE OF HPC AND AI
IN AGRICULTURE



JOHN DEERE



It started with a plow in **1837**

ERAS OF INNOVATION



1837-1852



1910-1918



1956-1964



1987-1997



TODAY

1982

Digital Tachometer with Clock

80C49

ROM: 2K

RAM: 128 B

~0 DMIPS

< 1 Watt



TODAY vPU2 Perception Controller

Orin AGX

1 TB
NVMem

64 GB
RAM

5.3 TFLOPS
(FP32)

275
TOPS (INT8)

100
Watts

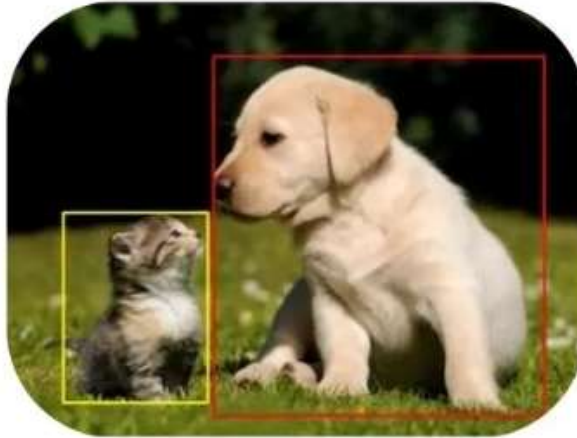
Machine Learning Vision Tasks

Is this a dog?



Image Classification

What is there in image
and where?



Object Detection

Which pixels belong to
which object?

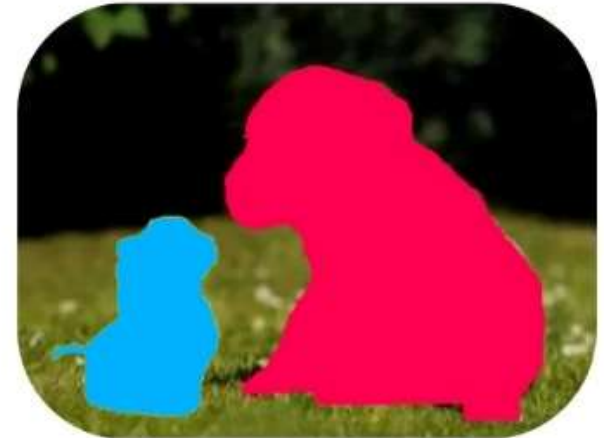


Image Segmentation

AUTOMATION

Using advanced technology to achieve **'super-human' performance.**

Increase profitability, productivity and sustainability for our customers.



AUTONOMY

Do all the above, but
without a driver in
the vehicle



SEE & SPRAY™



TECHNOLOGY

Computer Vision | Machine
Learning | Edge Computing
| Rugged cameras | Robotics







JOHN DEERE TECHNOLOGY

COMPUTER VISION ON COMBINES

HELP MENU







GPS

Cameras
(Back)

Cameras
(Front)

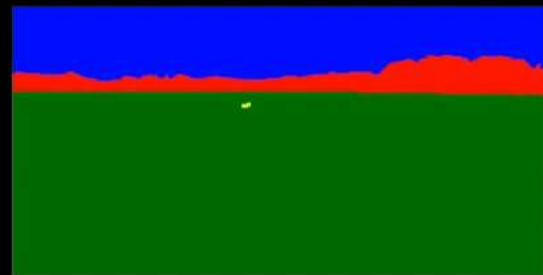
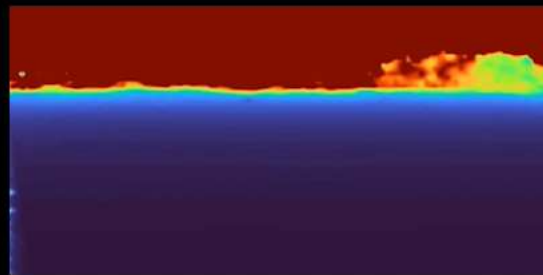
AI &
ML

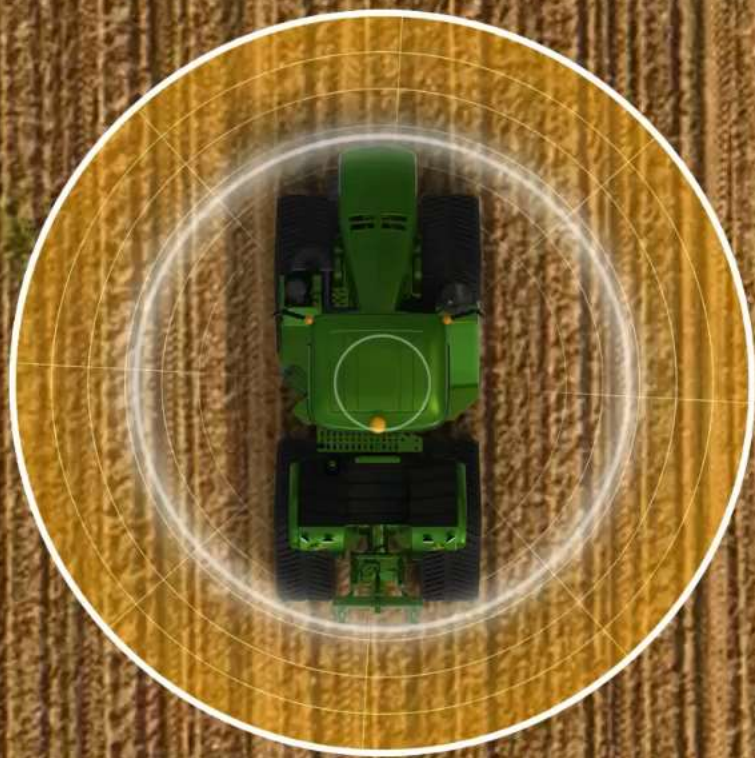
Sensors





What happens
when we
encounter an
OBSTACLE





CAMERA SYSTEM GEN 1
RANGE 16 m



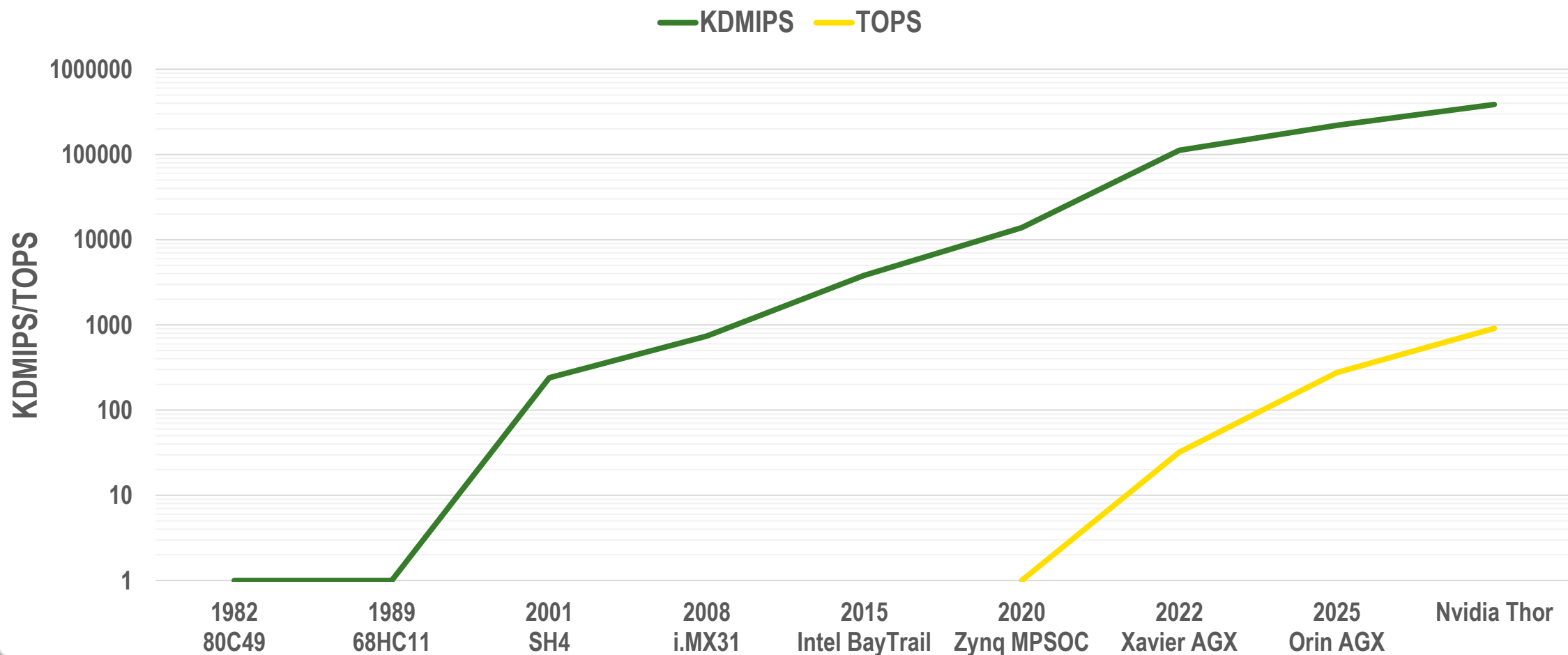
**High
Performance
Edge SoC**

**Unlocks
Automation &
Autonomy**

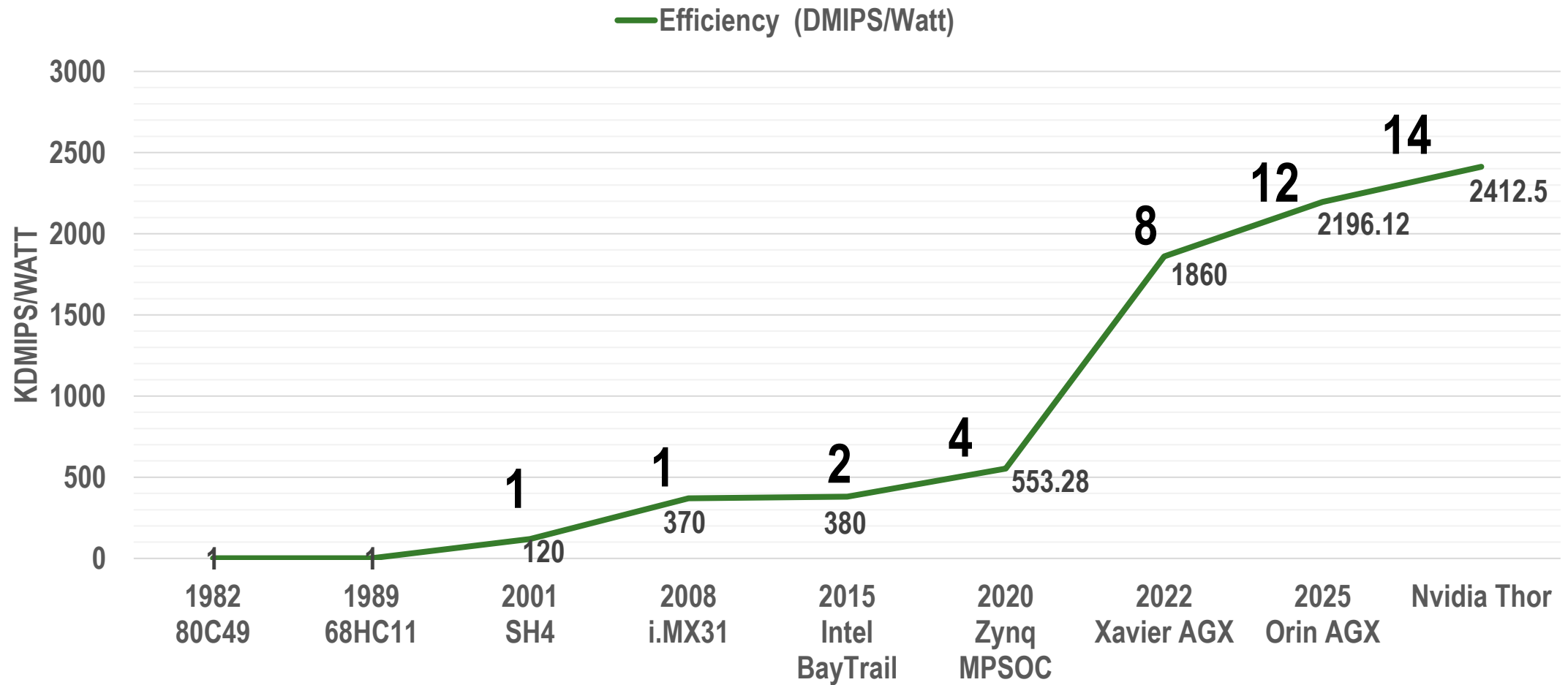
**High
Performance
Cameras**

**Advances in
Machine
Learning
Models**

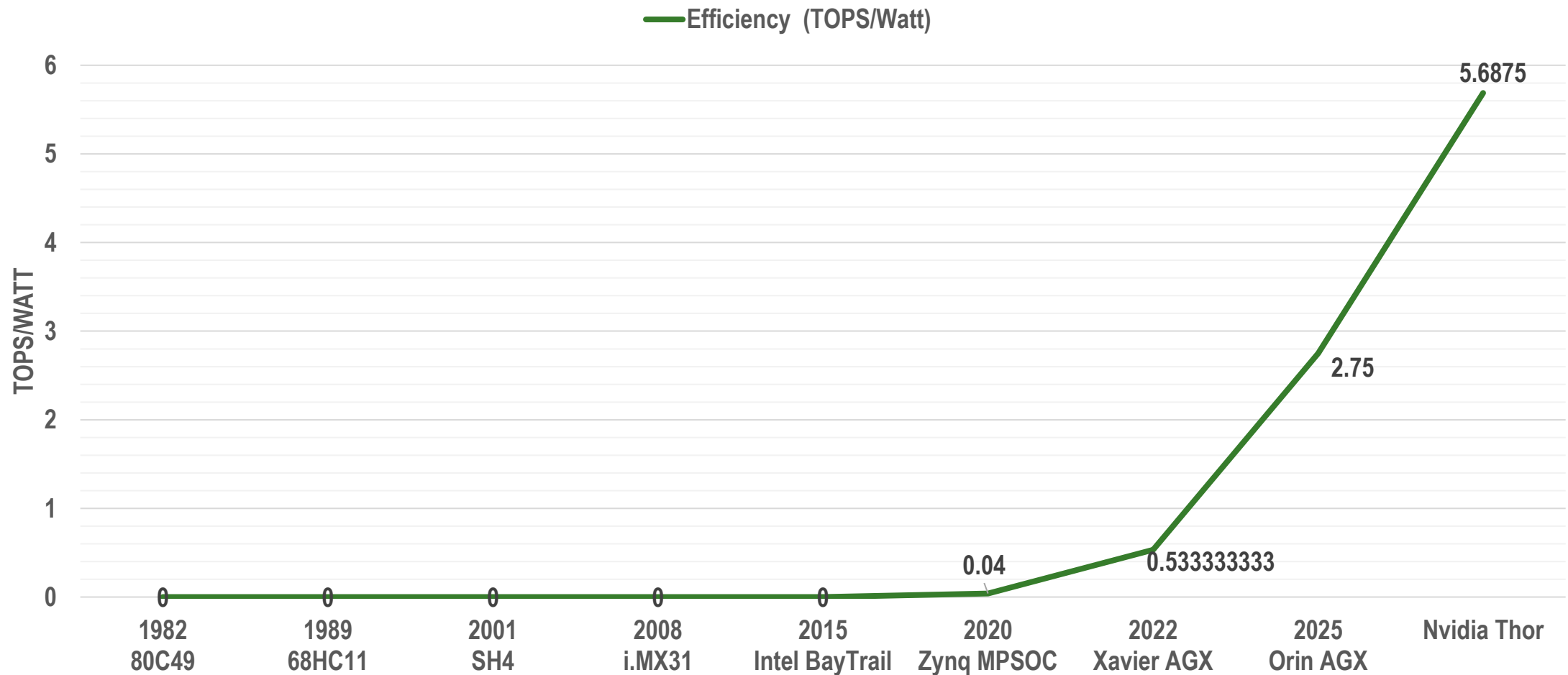
Rugged Edge Performance Over Time



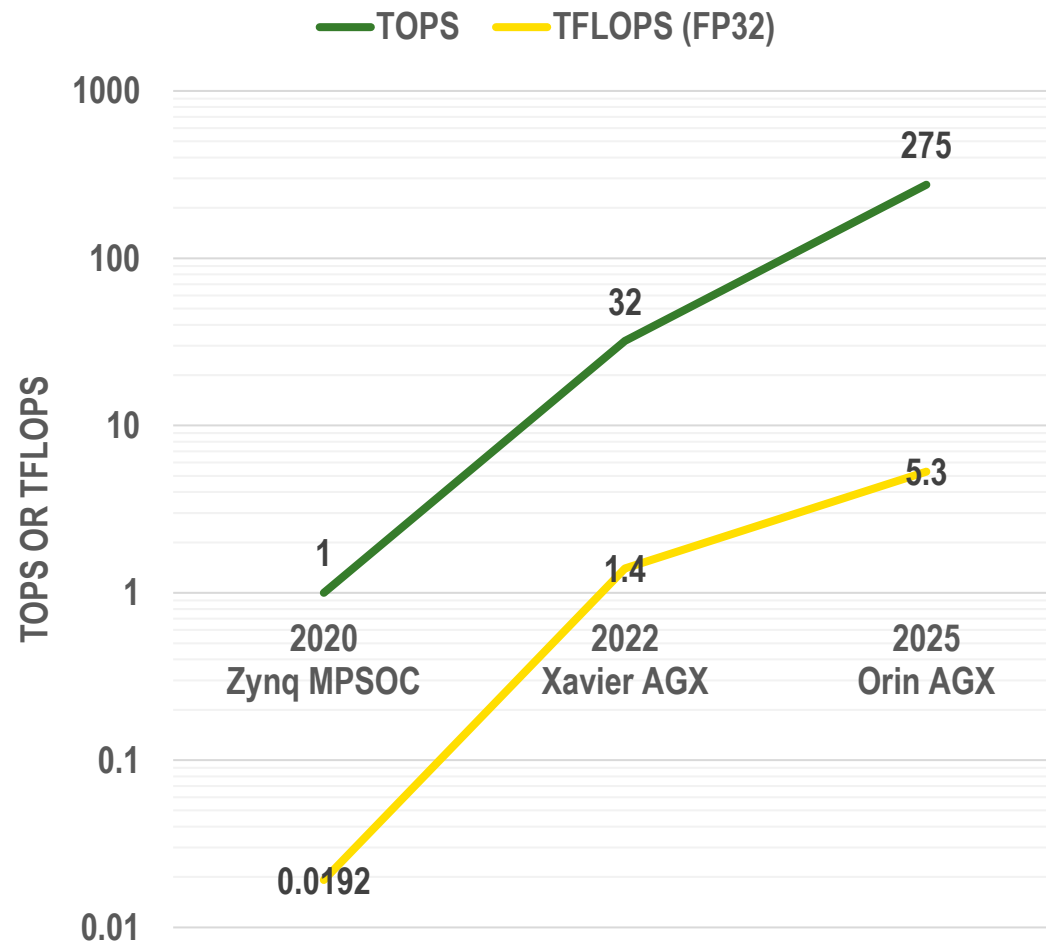
Efficiency of Rugged Edge Performance Over Time



Efficiency of Rugged Edge Performance Over Time

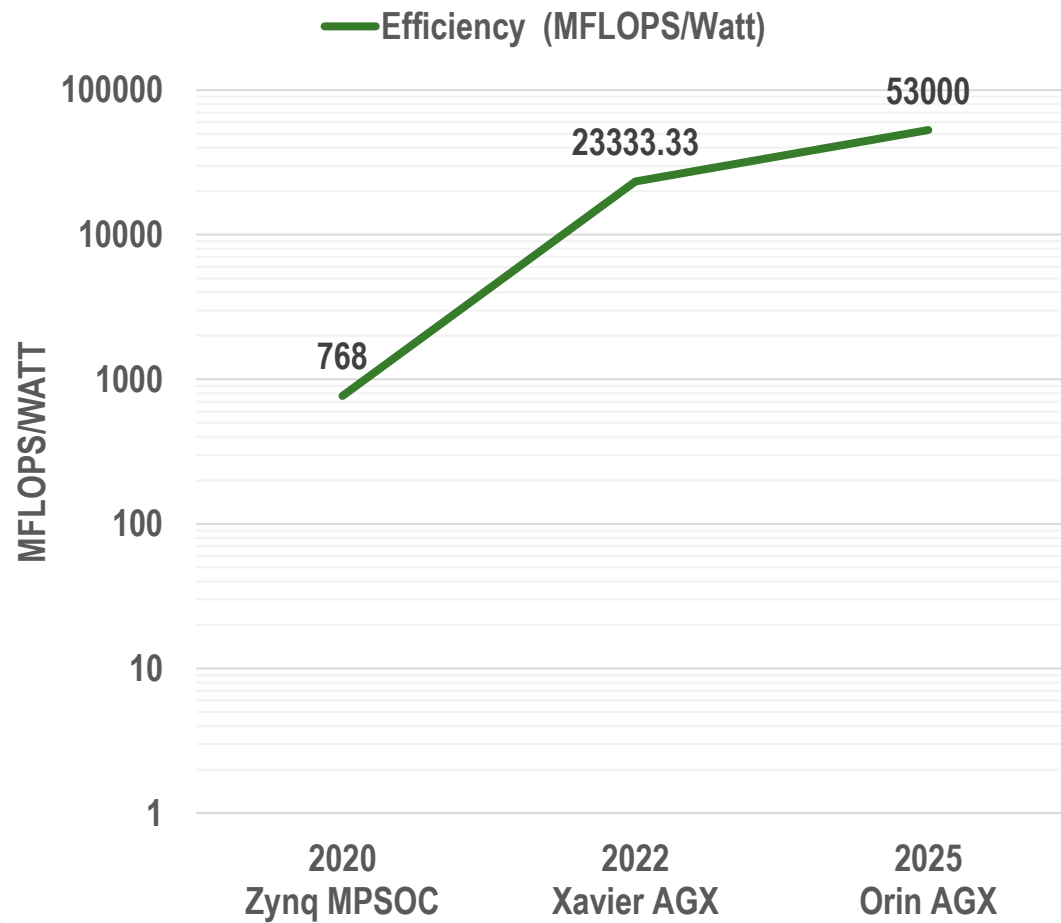


Top 500 Performance vs. Rugged Edge

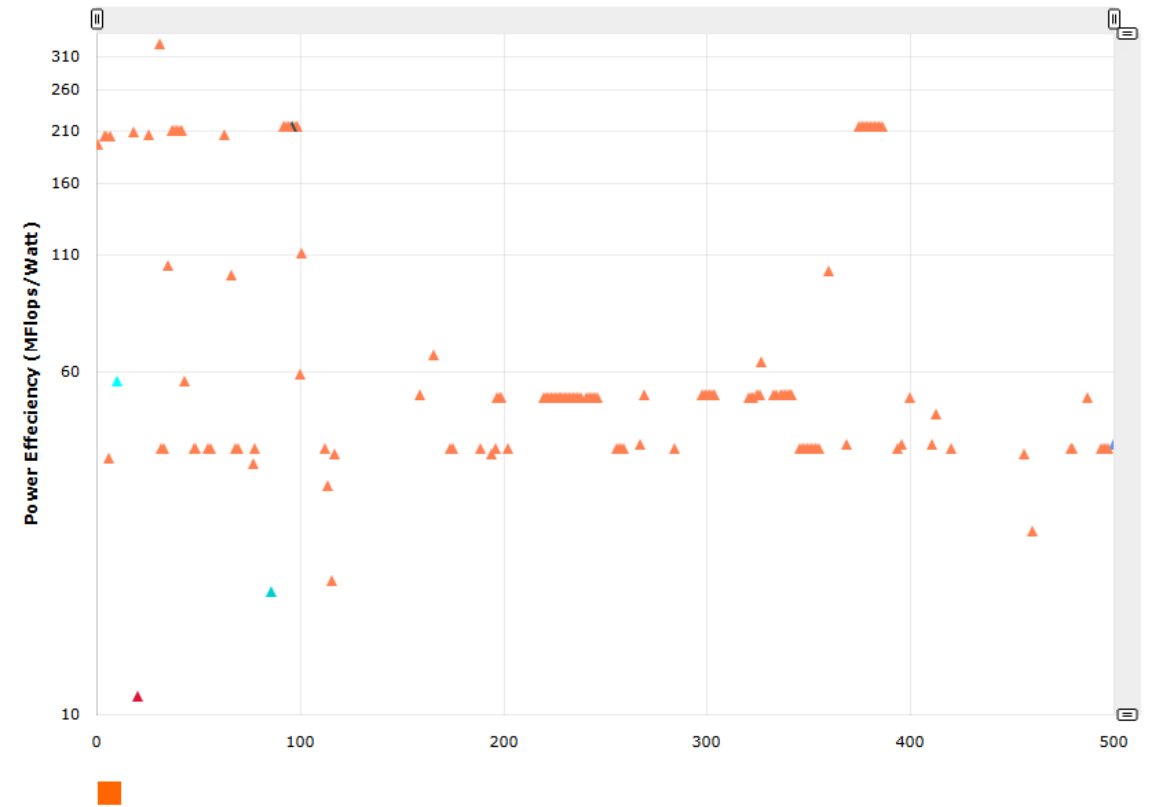


Source: top500.org

Top 500 Efficiency vs. Rugged Edge

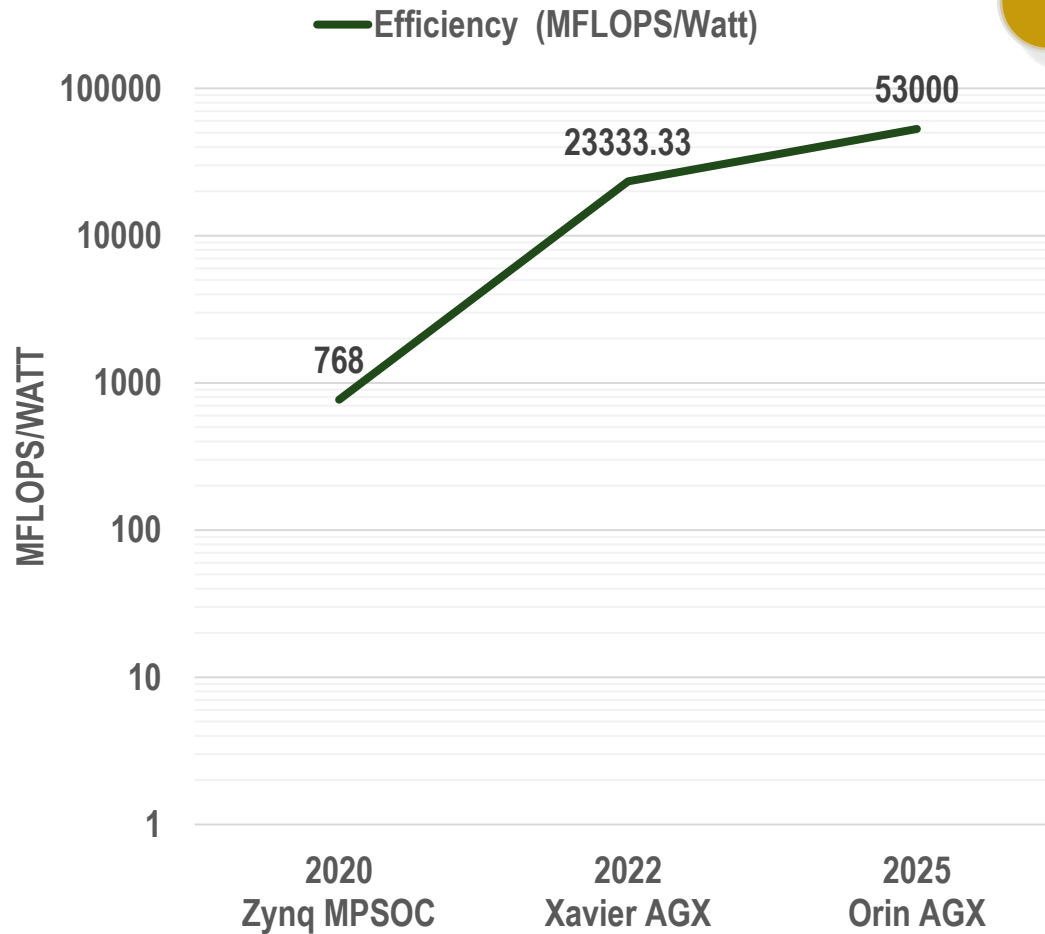


June 2007



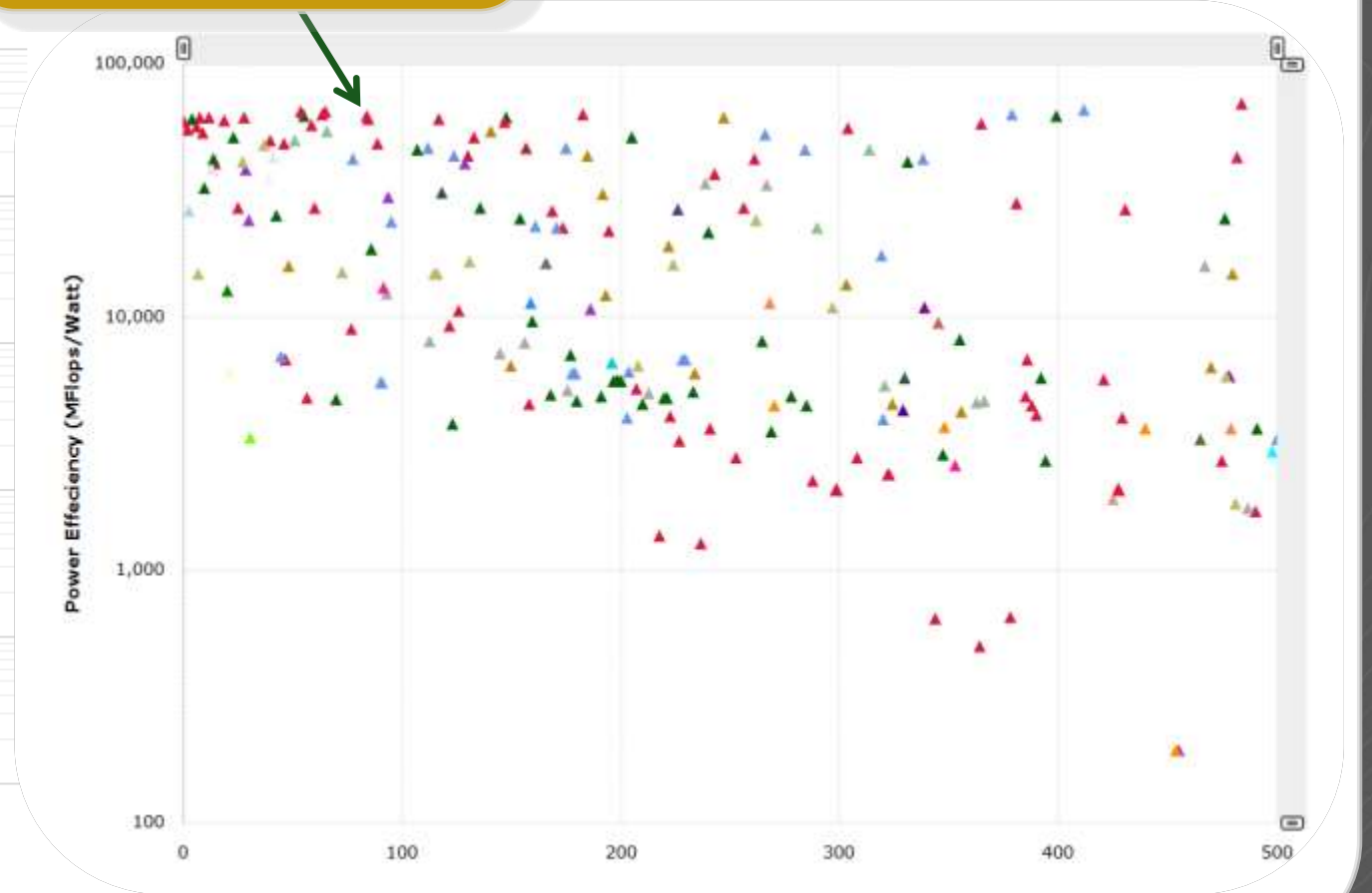
Source: top500.org

Top 500 Efficiency vs. Rugged Edge



64000 MFLOPS/Watt

June 2025

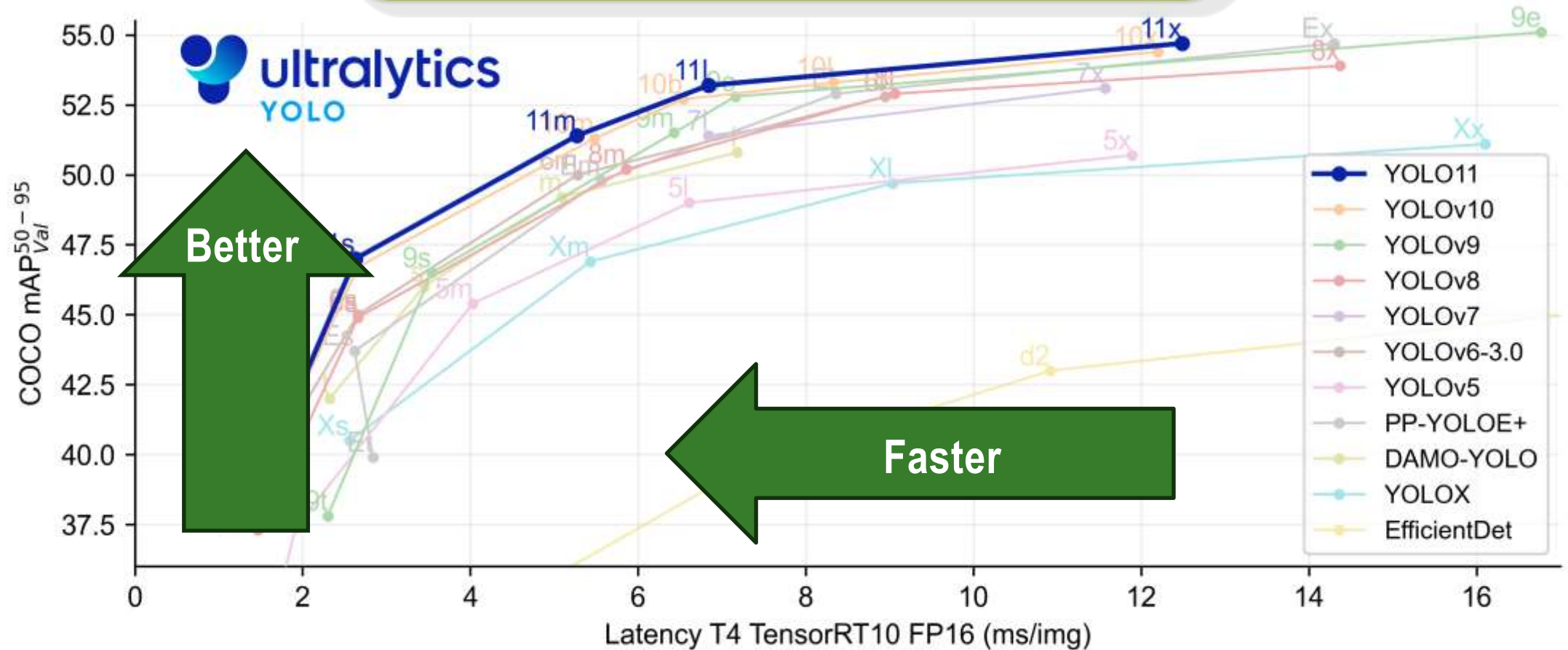


Source: top500.org

Machine Learning Over Time

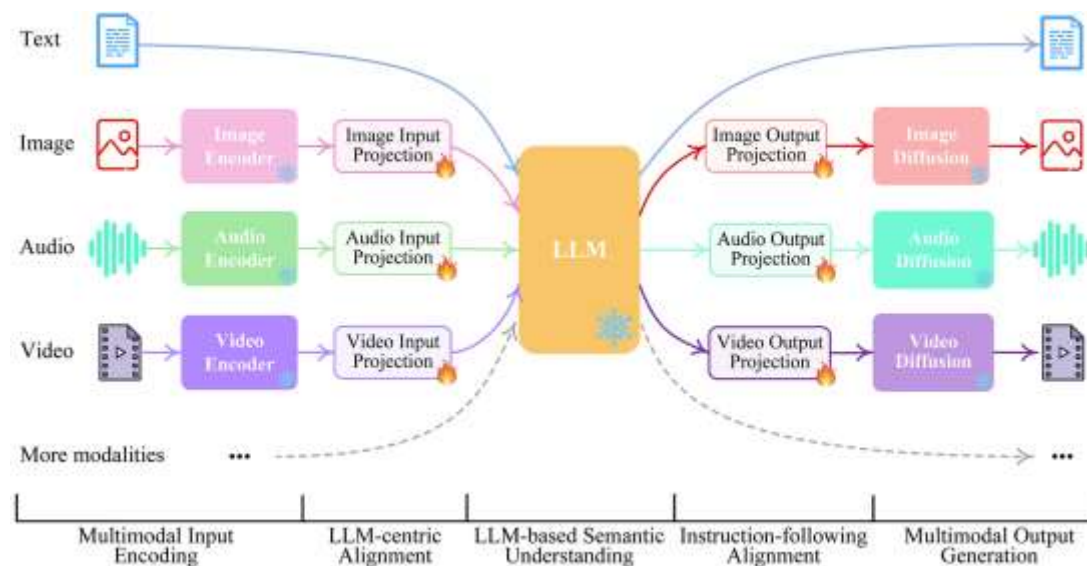
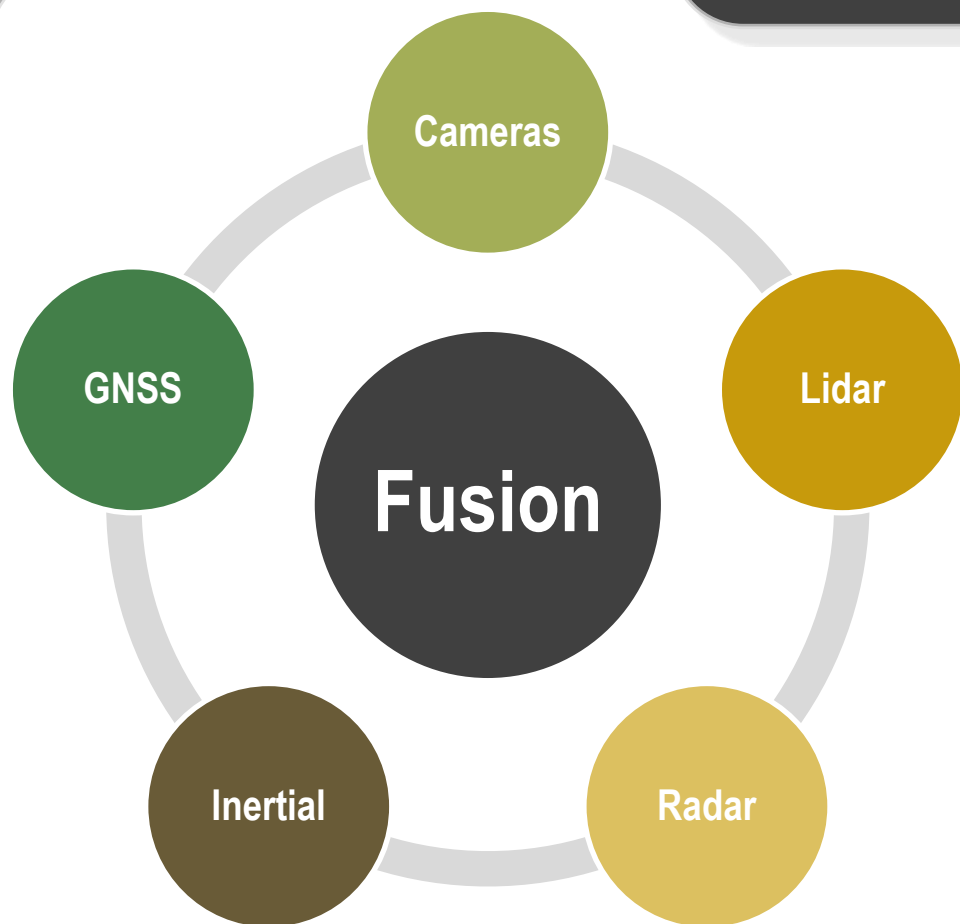
Model	Year	Parameters (M)	GFLOPS/Image	GLFOPS/Token
Resnet101 (classifier)	2015	43	7.83	NA
ConvNextV2 (classifier)	2023	15.6-660	7.21-600.8	NA
Faster RCNN (detector)	2015	41	134	NA
Yolo11m (detector)	2024	60.1	68	NA
DeeplabV3 (segmentation)	2015	61	258	NA
Dinov3 (segmentation)	2025	21-6716	63-14515	NA
Gemma3 (LLM)	2023	4300		476
Gemma3 (LLM)	2023	27000		3000
gpt-oss (LLM)	2025	20000		732

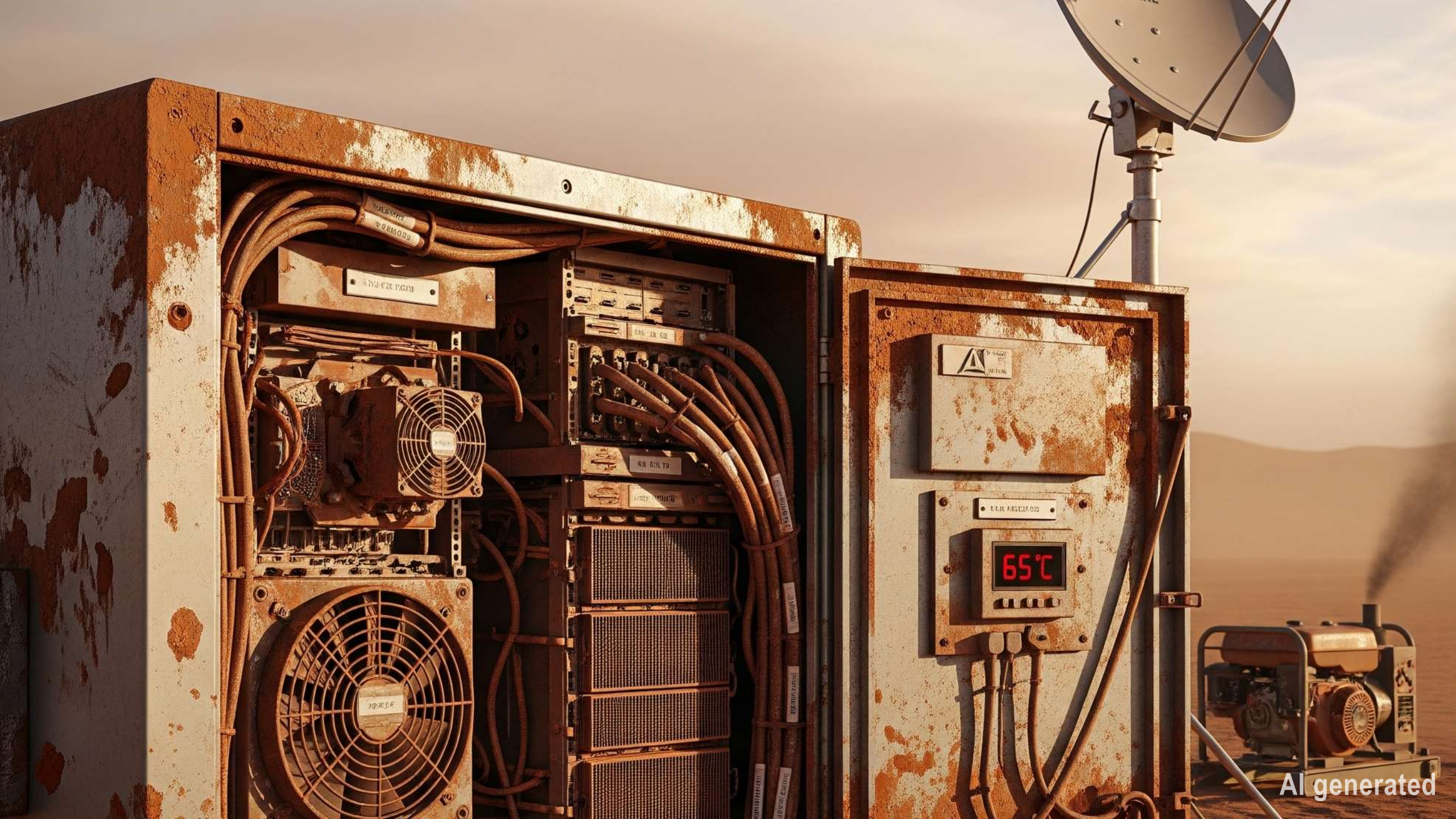
Object Detection Models



Source: <https://docs.ultralytics.com/models/yolo11>

Other Sensor Modalities





AI generated



Clean

**Limitless
power**

**Limitless
cooling**

**One of a
kind**

**Unlimited
budget**

Dirty... *really dirty*

Alternator or battery

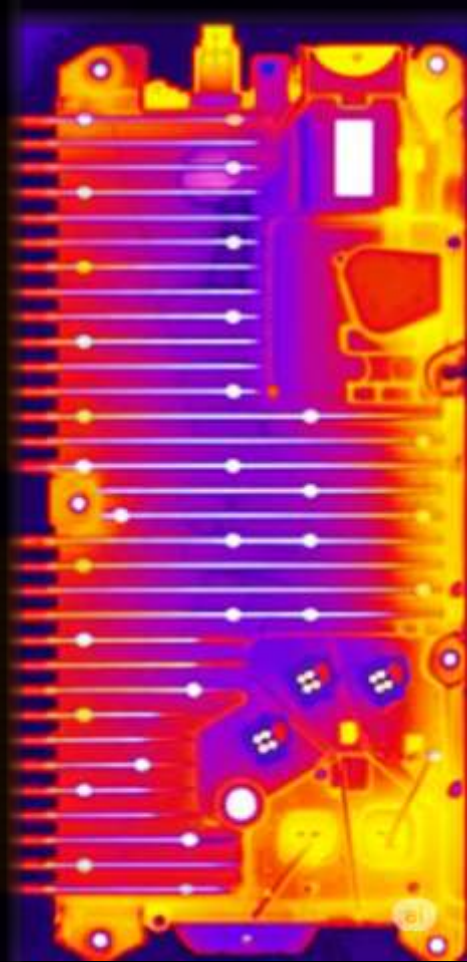
High vibration

**Passively cooled at high
temperatures**

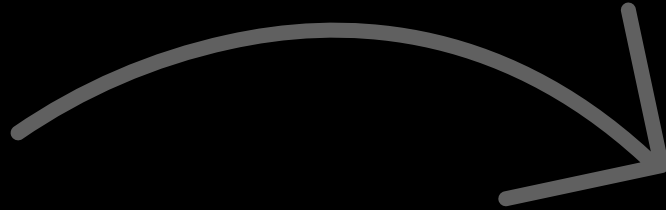
**Thousands of devices all
must work**

Controlling costs is critical





Performance/Watt is ***critically important*** for
passively cooled applications



We need more *efficient* AI models and not
simply **larger** ones



**We need new methods for sensor fusion and AI
(radar, lidar, vision, inertial, GNSS)**

AI generated



HPC and AI are enablers to
improve our customers productivity



JOHN DEERE

CONNECT
WITH ME ON
LINKEDIN

